The Great Equalizer: How Secondary Education Closes Wage Gaps in Malaysia

Introduction to Econometrics

Zalan Makany (Redacted), Lea Eggen (Redacted)

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1 Introduction

In this project, we dive into what actually drives wage rates for Malaysian workers. Instead of just assuming that things like age and education affect everyone's paycheck the exact same way, we built our econometric model step-by-step to see what the data really tells us. While we started out just looking at the basic returns to aging and total years of schooling, the real story turned out to be much more interesting. Our analysis highlights some pretty stark baseline wage gaps when it comes to gender and ethnicity. Here is the main takeaway: not all education is created equally. While primary school mostly just maintains these baseline inequalities, secondary education actually acts as a massive equalizer, actively closing the wage gaps between different demographic groups. The following sections walk through our process for exercises 1 to 4 to show exactly how we arrived at this conclusion.

2 Econometric Model and Methods

We use simple linear regressions and multivariate linear regressions. Consequently, our assumptions are the following.

Simple:

SLR1 (Linear in parameters)
Population model: $y = \beta_0 + \beta_1 x + u$

SLR2 (Random sampling)
We have a random sample of size n that follows the population model. (independent and identically distributed (i.i.d.) observations)

SLR3 (Sample variation in the explanatory variable)
Sample outcomes on x do not all have the same value.

SLR4 (Zero conditional mean)
$E(u|x) = 0$

SLR5 (Homoskedastic error terms)
$Var(u|x) = \sigma^2$

Multivariate:

MLR1 The model is $y_i = x_i'\beta + u_i$ with $x_i = (1, x_{i1}, ..., x_{ik})', i = 1, ..., n$

MLR2 $(x_i, y_i), i = 1, ..., n$ is a random sample from the population (independent and identically distributed (i.i.d) observation)

MLR3 No exact linear relationship between the regressors (no perfect collinearity or no (exact) multicollinearity)

MLR4 $E(u_i|x_i) = 0$ (zero conditional mean)

MLR5 $Var(u_i|x_i) = \sigma^2$ (homoskedastic error terms)

MLR6 All error terms are independent of X and of each other, and they all have the same distribution: $u|X \sim N_n(0, \sigma^2 I_n)$

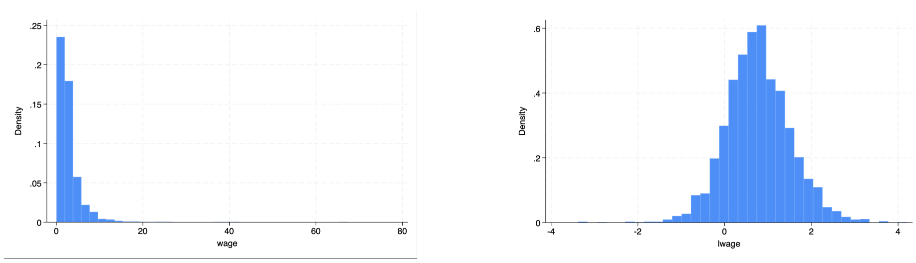
3 Data: Exercise 1

Based on our student numbers, we used the commands "drop if cat == 7" and "drop if urban == 0" to obtain our subsample of 1412 observations. The summary statistics reported in Table 1 show that age ranges from 15 to 65 years, which is a reasonable range for a working population. Our schooling variables range from 0 to 6 and 0 to 14, respectively. !!! Additionally we noticed a maximum wage of 67.24. At approximately 16 standard deviations above the mean, this initially appeared to be a massive outlier. However, wage distributions are inherently strongly right-skewed. So after taking the logarithm (which we can do since there is no data with wage = 0), using the command "generate lwage = ln(wage)", we managed to pull the extremities closer to the mean. (part a-d)

Variable	Obs	Mean	Std. dev.	Min	Max
man	1,412	.5630312	.4961869	0	1
malay	1,412	.3101983	.4627385	0	1
chinese	1,412	.4100567	.4920179	0	1
indian	1,412	.279745	.4490329	0	1
age	1,412	33.38031	10.61927	15	65
yprim	1,412	5.469547	1.487889	0	6
ysec	1,412	3.316572	2.83762	0	14
urban	1,412	1	0	1	1
cat	1,412	4.516997	2.987642	0	9
wage	1,412	3.446011	3.872129	.0553419	67.24037
lwage	1,412	.8944667	.8034252	-2.894225	4.208274

Table 1: Summary Statistics

The following two figures show the histograms of wage and ln(wage), respectively.



(a) Histogram of wage

(b) Histogram of lwage

Figure 1: Histograms of wage and log wage

From Figure 1a, we can see a strong right skew. This means more observations are clustered at lower wages, with very few being higher wages. From Figure 1b, we can see that taking the logarithm of the wages produces a more symmetric and bell-shaped curve, reducing the skewness and making the distribution more balanced.

We also looked at the structure of our other variables (Malay, Chinese, Indian, Man) in our subsample. We find that all take strictly 1 or 0 as values, so they are valid dummy variables. In addition, we checked whether our sample has anyone who is not one of the three included ethnicities or any multi-ethnic people.

```
count if (malay == 1) | (chinese==1) | (indian == 1)
  1,412
count if (malay == 1 & chinese==1) | ///
(indian == 1 & chinese==1) | (indian == 1 & malay==1)
  0
count if (malay == 0 & chinese==0 & indian == 0)
  0
```

The first check confirms that every individual belongs to at least one ethnicity. The second shows that no individual belongs to two or more groups simultaneously. The third confirms that no individual is unclassified. Therefore, the three dummies are mutually exclusive and collectively exhaustive, implying

$$\text{malay} + \text{chinese} + \text{indian} = 1$$

for each observation. This linear relationship means that we cannot include all three dummies in a regression with a constant term without creating perfect multicollinearity (violation of assumption MLR3). In our subsequent models, as such we include only two of the dummies and treat the third as the base category. (part e)

4 Results

4.1 Exercise 2

Then, we started with the simplest possible model that attempted to explain wage differences with only total years of schooling, for the moment assuming that all education provides equal value, which we will relax later. For this we created a new variable "generate $yschool = yprim + ysec$". As such, the model looks the following: $y = \beta_0 + \beta_1 yschool + u$. (Using assumptions SLR1-SLR4)

Source	SS	df	MS			
				Number of obs	=	1,412
				F(1, 1410)	=	460.10
Model	224.079639	1	224.079639	Prob > F	=	0.0000
Residual	686.709544	1,410	.487028046	R-squared	=	0.2460
				Adj R-squared	=	0.2455
Total	910.789184	1,411	.64549198	Root MSE	=	.69787

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yschool	.1073256	.0050036	21.45	0.000	.0975104	.1171408
_cons	-.0485088	.0477239	-1.02	0.310	-.1421263	.0451086

Table 2: ln(wage) and schooling

From the intercept, $\hat{\beta}_0$, we can see that for an individual with zero years of schooling, the predicted wage is about 0.95 Malaysian dollars (calculated as $e^{-0.0485}$). Note that the negative coefficient here means the predicted wage is less than 1.

From the slope, $\hat{\beta}_1$, we can see that for every additional year of schooling, the predicted wage is expected to increase by approximately 10.73%.

For part g, Figure 2 contains the observations($yschool$, $lwage$) and all the predictions of the model in the previous question.

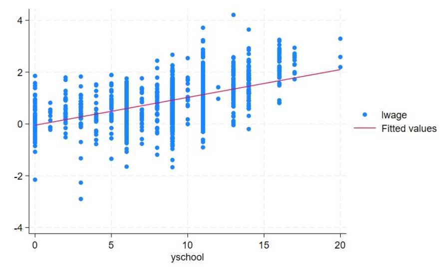


Figure 2: $yschool$ and $lwage$ scatter plot

When interpreting the estimated slope coefficient as the causal effect of yschool on lwage, we need to rely on various assumptions for a simple linear regression. However there are rather serious threats to these assumptions (part h)

- **SLR1** (linearity in parameters): if the true relationship is nonlinear, the estimated coefficient may represent a misleading average effect.
- **SLR2** (random sampling): if the sample is not random, results may suffer from selection bias and not generalize.
- **SLR3** (sample variation in yschool): if there is no variation in schooling, the slope cannot be estimated at all.

  And most importantly:

- **SLR4** - the zero conditional mean condition $E(u|\ yschool) = 0$. This means that, on average, unobserved factors affecting wages (ability, motivation, family background) are uncorrelated to years of schooling.
  When this fails, the OLS slope is biased and cannot be interpreted as causal. The three most common violations are:
  - **Measurement Error**
  - **Reverse causaility:** if say higher wages attract people to stay in school for longer then the direction of causality isn't clear
  - **Omitted variable bias:** probably the most grave issue, if there's a variable that's not included in the model, but correlates to both the years of schooling and wages for example motivation, family background, discrimination then that could bias our estimation

So for this model it would be difficult to argue causality. However, in the following chapters we will continue to refine the model and loosen some assumptions.

For part i, we test the claim that the coefficient of *yschool* is larger than 1 using the following statistical test:

```
. display ( _b[yschool] -1) / _se[yschool]
-178.40784

. display invttail(1410, 0.05)
1.645935
```

Figure 3: Statistical test

In Figure 3, we compute the T statistic and the critical value of the t distribution using the null hypothesis, $H_0 : \beta_1 \leq 1$ and the alternative hypothesis, $H_1 : \beta_1 > 1$ with a one-sided right tailed t-test. As we can see, after calculating the critical value using our degrees of freedom we get $t_{critical} = 1.645935$ which is larger than our T statistic $T = -178.40784$. Therefore, we fail to reject the null hypothesis at the 95% level. We can also perform a left-tailed test with $H_0 : \beta_1 \geq 1$ and $H_1 : \beta_1 < 1$. Here $t_{critical} = -1.645935$ and $T = -178.40784$, so we reject this null at the 95% level. This provides direct evidence that β_1 is less than 1, contradicting our colleague's hypothesis. (using SLR1 - SLR5)

4.2 Exercise 3

The following table is for part a, which is the regression model explaining lwage from age, yschool, man and the ethnicity dummies.

Source	SS	df	MS			
Model	435.120402	5	87.0240803	Number of obs	=	1,412
Residual	475.668782	1,406	.338313501	F(5, 1406)	=	257.23
Total	910.789184	1,411	.64549198	Prob > F	=	0.0000
				R-squared	=	0.4777
				Adj R-squared	=	0.4759
				Root MSE	=	.58165

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yschool	.1314734	.0044088	29.82	0.000	.1228248	.1401219
age	.0311518	.0015389	20.24	0.000	.0281329	.0341707
man	.2615843	.0318038	8.22	0.000	.1991963	.3239724
malay	.1155468	.041086	2.81	0.005	.0349504	.1961432
chinese	.2934227	.0379775	7.73	0.000	.218924	.3679215
_cons	-1.603974	.0753564	-21.29	0.000	-1.751797	-1.456151

Table 3: Multivariate linear regression

The increase of R-squared is expected as there are more explanatory variables. To check whether this increase is substantial and whether it is worth it to add these additional variables, we look at the Adjusted R-squared, which in this case also increased. Therefore, we can conclude that it is worth adding these additional explanatory variables. (Using assumptions MLR1-MLR4)

To check individual significance of these added variables, we can look at the P values under associated with the t statistics of each regressor. We can see that for all variables the p values are less than $\alpha = 0.05$. This allows us to reject the null $H_0 : \beta_j = 0$. In favour of the alternative hypothesis $H_1 : \beta_j \neq 0$. Meaning age, years of schooling, gender, and ethnicity all have a statistically significant relationship with log-wages at the 95% confidence level. (Here using MLR5)

For part b, we are checking the model F-test,

H_0 : all $\beta = 0$

H_1 : at least 1 β is not equal to 0

Using the F-test table, we can determine that the critical value is f = 2.21. Looking at the F-statistic : $F(5, 1406) = 257.23$, we can see that it is significantly higher than the critical value, so, our model has statistically significant explanatory power and we reject H_0. Looking at the MP$Prob > F = 0.0000$, we can see that the p-value is less than $\alpha = 0.05$ so we reject H_0.

For part c, we use an F test on our ethnicity dummies,

H_0 : both β of the ethnicity dummies is 0

H_1 ; at least one β is not 0

```
. test malay chinese

 ( 1)  malay = 0
 ( 2)  chinese = 0

       F(  2,  1406) =   31.39
            Prob > F =    0.0000
```

Figure 4: F-test for part c

Taking $Prob > F = 0.0000$, we can see that the p-value is less than $\alpha = 0.05$ so we therefore, reject H_0.

For part d, we constructed a 90% confidence interval for the coefficient of Age using Stata as followed:

lwage	Coefficient	Std. err.	t	P>|t|	[90% conf.	interval]
(1)	.0311518	.0015389	20.24	0.000	.0286188	.0336848

Figure 5: 90% confidence interval

For part e, from Figure 6, we can see that both ysec and yprim are, as expected, statistically significant. And that the explanatory power of the model has increased by looking at the Adj R-squared. We can see that secondary education has a greater estimated coefficient than primary education, meaning there might be greater returns to secondary education.

To see whether there is a point in looking at years of schooling separately we need to check whether the effect of one is significantly different from the other. To do this we ran a t-test on $\beta_{ysec} = \beta_{yprim}$ or equivalently $\beta_{ysec} - \beta_{yprim} = 0$

. regress lwage ysec yprim age man malay chinese

Source	SS	df	MS			
				Number of obs	=	1,412
				F(6, 1405)	=	222.32
Model	443.577489	6	73.9295815	Prob > F	=	0.0000
Residual	467.211694	1,405	.332535014	R-squared	=	0.4870
				Adj R-squared	=	0.4848
Total	910.789184	1,411	.64549198	Root MSE	=	.57666

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ysec	.1526899	.0060667	25.17	0.000	.1407891	.1645908
yprim	.0740347	.0121996	6.07	0.000	.0501032	.0979661
age	.0289857	.0015851	18.29	0.000	.0258764	.032095
man	.2907996	.0320588	9.07	0.000	.2279112	.3536879
malay	.113842	.040735	2.79	0.005	.0339341	.1937499
chinese	.3032263	.037702	8.04	0.000	.2292681	.3771845
_cons	-1.307812	.0950287	-13.76	0.000	-1.494225	-1.121398

Figure 6: ysec yprim regression

```
 ( 1)  ysec - yprim = 0

       F(  1,  1405) =    25.43
            Prob > F =    0.0000
```

Figure 7: F-test

For part f, Figure 8 shows the new regression with the new variable agesq (“`generate agesq = age^2`”).

Source	SS	df	MS			
				Number of obs	=	1,412
				F(6, 1405)	=	247.04
Model	467.574591	6	77.9290984	Prob > F	=	0.0000
Residual	443.214593	1,405	.315455226	R-squared	=	0.5134
				Adj R-squared	=	0.5113
Total	910.789184	1,411	.64549198	Root MSE	=	.56165

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
age	.1180401	.0086943	13.58	0.000	.1009849	.1350952
agesq	-.0011883	.0001172	-10.14	0.000	-.0014181	-.0009585
yschool	.1219256	.0043601	27.96	0.000	.1133726	.1304786
man	.2549262	.0307176	8.30	0.000	.1946688	.3151836
malay	.0793565	.0398338	1.99	0.047	.0012163	.1574967
chinese	.2893424	.0366743	7.89	0.000	.2174	.3612847
_cons	-2.945821	.1509845	-19.51	0.000	-3.242	-2.649642

Figure 8: agesq regression

The coefficients of age and agesq suggest that initially there are positive returns to aging of approximately 11.8%. however since $\hat{\beta}_{agesq}$ is negative and significant we can be confident in observing decreasing returns and later on even potentially negative returns. To check if this is the case we chose to calculate where the effect of age is maximal. Aka where the derivative of the second degree polinomial is 0. $age_{max} = -\frac{\hat{\beta}_{age}}{2\hat{\beta}_{agesq}} \approx 49.67$ We observe that until age 50 there are positive returns, however after we get negative returns to aging. (We use assumptions MLR1-MLR4)

To check whether there is a significant difference between the learning of man and women we created a new regression with an additional interaction term. From Table 4, we can see that the p-value $P_{i.man\#c.yschool} = 0.284$ is greater than $\alpha = 0.05$, therefore, we do not reject $H_0 : \beta_{i.man\#c.yschool} = 0$.
(additionally using MLR5)

. regress lwage yschool age man malay chinese c.man#c.yschool

Source	SS	df	MS			
				Number of obs	=	1,412
				F(6, 1405)	=	214.57
Model	435.508748	6	72.5847913	Prob > F	=	0.0000
Residual	475.280436	1,405	.33827789	R-squared	=	0.4782
				Adj R-squared	=	0.4759
Total	910.789184	1,411	.64549198	Root MSE	=	.58162

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yschool	.1360766	.0061557	22.11	0.000	.1240011	.148152
age	.0312545	.0015418	20.27	0.000	.0282299	.0342791
man	.339872	.0796878	4.27	0.000	.1835521	.496192
malay	.1154169	.041084	2.81	0.005	.0348243	.1960094
chinese	.2934112	.0379755	7.73	0.000	.2189163	.3679061
c.man#c.yschool	-.0089538	.0083567	-1.07	0.284	-.0253467	.0074391
_cons	-1.647554	.0856291	-19.24	0.000	-1.815529	-1.479579

Table 4: Effect of schooling on men and women

4.3 Exercise 4

Initially, we considered building our model by simply including all variables that appeared statistically significant. However, this approach, often referred to as 'data mining' [Wooldridge, 2019] can artificially inflate the R^2 and result in unreliable inferences. To avoid this, we adopted a more theory driven strategy: we estimated a carefully selected set of candidate models based on economic reasoning and compared them using the adjusted R^2, which appropriately penalizes the inclusion of irrelevant variables. The following section details our preferred specification.

We started by declaring a baseline using what we saw in the previous section. Of the form:

$$y = \beta_0 + \beta_1 ysec + \beta_2 yprim + \beta_3 age + \beta_4 agesq + \beta_5 man + \beta_6 malay + \beta_7 chinese + u$$

When we include the different levels of education separately, alongside the second-order term for age, we find an interesting result. In our previous, more restricted models, being Malay showed a significant wage difference compared to the Indian baseline. However, in this full model, that difference is no longer statistically significant. This allows us to make some clear demographic inferences about our dataset.

Source	SS	df	MS
Model	479.445646	7	68.4922351
Residual	431.343538	1,404	.307224742
Total	910.789184	1,411	.64549198

Number of obs	=	1,412
F(7, 1404)	=	222.94
Prob > F	=	0.0000
R-squared	=	0.5264
Adj R-squared	=	0.5240
Root MSE	=	.55428

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ysec	.1466368	.0058581	25.03	0.000	.1351451	.1581284
yprim	.0530534	.0118859	4.46	0.000	.0297375	.0763694
age	.1203057	.0085878	14.01	0.000	.1034594	.1371521
agesq	-.0012545	.0001161	-10.81	0.000	-.0014823	-.0010268
man	.2893151	.0308149	9.39	0.000	.2288668	.3497634
malay	-.2254681	.0359305	-6.28	0.000	-.2959514	-.1549848
indian	-.3007791	.0362395	-8.30	0.000	-.3718685	-.2296898
_cons	-2.36746	.1561151	-15.16	0.000	-2.673704	-2.061216

Figure 9: Baseline

First, the Malay workers in our sample are likely younger than the Indian workers, meaning the baseline model was absorbing the negative wage effects of aging into the Malay variable. Second, they are likely to have a higher proportion of secondary education. Because secondary schooling yields roughly three times the wage premium of primary schooling, our previous model, which combined these schooling years, artificially inflated the ethnicity dummy to account for that hidden educational advantage. This serves as a textbook example of omitted variable bias. Since Malay is no longer significant after these controls, we drop it from the rest of our models. However the Chinese dummy remains significant and in an auxiliary regression not included here we found that there is also a significant difference between Chinese and Malay workers. So we decided to keep only the Chinese dummy and continue to investigate differences between Chinese and non-Chinese workers.

Next, we wondered how these ethnic differences change with years of schooling, so we ran a new regression with two additional interaction variables:

$$y = \beta_0 + \beta_1 ysec + \beta_2 yprim + \beta_3 age + \beta_4 agesq + \beta_5 man + \beta_6 chinese + \\ + \beta_7 chinese \times yprim + \beta_8 chinese \times ysec + u$$

This again is quite interesting. From $\beta_{chinese} \approx 0.365$ we can see that Chinese workers with 0 years of schooling start with an approximately 36.5% wage premium, which seems to increase at the same rate as for others with each additional year of primary school education. However, Non-Chinese workers seem to get greater benefits from secondary school.

Finally with this more sophisticated model we decided to once again check for the effect of schooling on gender differences.

Source	SS	df	MS			
Model	480.357239	8	60.0446549	Number of obs	=	1,412
Residual	430.431944	1,403	.306793973	F(8, 1403)	=	195.72
				Prob > F	=	0.0000
Total	910.789184	1,411	.64549198	R-squared	=	0.5274
				Adj R-squared	=	0.5247
				Root MSE	=	.55389

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yprim	.0552645	.0151495	3.65	0.000	.0255464	.0849825
ysec	.158431	.0074475	21.27	0.000	.1438216	.1730404
age	.1223761	.0085533	14.31	0.000	.1055975	.1391547
agesq	-.0012806	.0001157	-11.07	0.000	-.0015075	-.0010537
man	.2913317	.0308887	9.43	0.000	.2307386	.3519247
chinese	.3646197	.1145201	3.18	0.001	.1399706	.5892688
c.chinese#c.yprim	-.0029596	.0221271	-0.13	0.894	-.0463653	.0404461
c.chinese#c.ysec	-.0269249	.0117855	-2.28	0.022	-.0500441	-.0038057
_cons	-2.721447	.1605642	-16.95	0.000	-3.036419	-2.406475

Figure 10: Chinese Interactions

$$y = \beta_0 + \beta_1 ysec + \beta_2 yprim + \beta_3 age + \beta_4 agesq + \beta_5 man + \beta_6 chinese + \\ + \beta_7 chinese \times ysec + \beta_8 man \times yprim + \beta_9 man \times ysec + u$$

This new model yields remarkably different results compared to our earlier, simpler iterations. We observe a trend mirroring the ethnic wage differences: men begin with a substantial wage premium at zero years of schooling of around 53.24%. While this gap remains parallel and undiminished throughout the primary schooling years. However, every subsequent year of secondary education significantly closes the divide by approximately 2.78%.

Source	SS	df	MS			
Model	483.879914	9	53.7644349	Number of obs	=	1,412
Residual	426.909269	1,402	.304500192	F(9, 1402)	=	176.57
				Prob > F	=	0.0000
Total	910.789184	1,411	.64549198	R-squared	=	0.5313
				Adj R-squared	=	0.5283
				Root MSE	=	.55182

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yprim	.061269	.0159353	3.84	0.000	.0300094	.0925287
ysec	.1762786	.0100206	17.59	0.000	.1566215	.1959357
age	.1255564	.0085682	14.65	0.000	.1087486	.1423643
agesq	-.0013229	.0001159	-11.42	0.000	-.0015502	-.0010956
man	.5323947	.1153943	4.61	0.000	.3060305	.7587589
chinese	.3629358	.0457917	7.93	0.000	.2731081	.4527635
c.chinese#c.ysec	-.0309043	.0107311	-2.88	0.004	-.0519551	-.0098535
c.man#c.yprim	-.0270029	.0224076	-1.21	0.228	-.0709589	.016953
c.man#c.ysec	-.0278211	.0117022	-2.38	0.018	-.0507767	-.0048655
_cons	-2.869431	.1694609	-16.93	0.000	-3.201855	-2.537007

Figure 11: Gender Interactions

Now by dropping the statistically insignificant interaction between gender and years of primary schooling we created our final model.

$$y = \beta_0 + \beta_1 ysec + \beta_2 yprim + \beta_3 age + \beta_4 agesq + \beta_5 man + \beta_6 chinese + \\ + \beta_7 chinese \times ysec + \beta_8 man \times ysec + u$$

It is important to note that we technically have a decrease in explanatory power when looking at the adjusted R^2 of our model of about 0.0002. But we believe this decrease is marginal enough to make using the simpler model the better option. (part a)

Source	SS	df	MS
Model	483.437713	8	60.4297141
Residual	427.35147	1,403	.30459834
Total	910.789184	1,411	.64549198

Number of obs	=	1,412
F(8, 1403)	=	198.39
Prob > F	=	0.0000
R-squared	=	0.5308
Adj R-squared	=	0.5281
Root MSE	=	.5519

lwage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yprim	.0485618	.0119495	4.06	0.000	.025121	.0720027
ysec	.1794811	.0096634	18.57	0.000	.1605248	.1984373
age	.124684	.0085389	14.60	0.000	.1079336	.1414345
agesq	-.0013131	.0001156	-11.36	0.000	-.0015398	-.0010863
man	.4055363	.0472746	8.58	0.000	.3127998	.4982728
chinese	.3567703	.0455124	7.84	0.000	.2674907	.44605
c.chinese#c.ysec	-.0298905	.0106998	-2.79	0.005	-.0508799	-.0089012
c.man#c.ysec	-.0337729	.0106105	-3.18	0.001	-.0545872	-.0129587
_cons	-2.794967	.1578188	-17.71	0.000	-3.104553	-2.485381

Figure 12: Final model

We used this model to construct the 80% prediction interval for the wage of a woman of Malay ethnicity (aged 30, with 6 years of primary schooling and no secondary schooling), we first generated a new observation containing these specific parameters. Since Malay is not significantly different from the base group after controlling for age and schooling, our model treats a Malay worker as an Indian worker. Using the predict command, we calculated both the predicted log wage ($\hat{y}$) and the standard error of the prediction (se_{pred}). Finally, we calculated the interval bounds using the formula $[\hat{y} - se_{pred} \times t^*, \hat{y} + se_{pred} \times t^*]$, where t^* represents the critical value of the t-distribution for the given degrees of freedom. Important to note that because the model estimates the log wage, this interval is currently in log points, to express the interval in actual Malaysian dollars, the lower and upper bounds must be exponentiated $[e^{\hat{y} - se_{pred} \times t^*}, e^{\hat{y} + se_{pred} \times t^*}]$. This process gives the following final result:

```
.
. display "Lower Bound:" exp(y_pred[1413] - se_pred[1413] * cr)
Lower Bound:.51947806

. display "Upper Bound:" exp(y_pred[1413] + se_pred[1413] * cr)
Upper Bound:2.1495146
```

Figure 13: Prediction

As we can see in Figure 13 our model predicts with 80% confidence that the wage of such a person would be between \$0.519 and \$2.15 -s. (part b)

5 Conclusion

Looking back at our modeling process, it's clear how easy it is to get misled by omitted variable bias. By moving away from our initial, overly simple regressions and building a more thoughtful multivariate model with interaction terms, we were able to uncover the real mechanics behind the wages in our sample.

Three main takeaways stand out. First, getting older doesn't just keep bumping up your pay forever, the returns to aging actually max out around age 50 and then start to drop off. Second, primary and secondary education shouldn't just be lumped together as "years of schooling" as secondary school provides a significantly bigger boost to wages.

Finally, and most importantly, our model highlights just how powerful secondary education is for leveling the playing field. Men in our sample start out with a massive 53.24% wage premium, and Chinese workers start with a 36.5% premium over their peers. Going through primary school doesn't really change these parallel gaps. But the moment workers hit secondary education, the dynamic shifts. Every extra year of secondary school actively closes the gender wage gap by nearly 2.8% and starts tipping the scales in favor of non-Chinese workers. In the end, our analysis shows that for this dataset, secondary school isn't just about making more money, but it's a key driver for closing systemic wage divides.

5.1 Limitations

While our final model is a big step up from our initial regressions, we still need to take these results with a slight grain of salt. Because we are working with observational data rather than a controlled experiment, it is incredibly difficult to claim strict causality.

Even though we controlled for key demographic and educational factors, there are definitely still unobserved variables hiding in our error term, u. Things like an individual's natural ability, personal motivation, family connections, or the specific industry they work in are simply not captured in our dataset. If these missing factors correlate with both a person's wages and their level of education, we might still be dealing with some lingering omitted variable bias.

Additionally, our education variables (yprim and ysec) only measure the quantity of schooling, not the quality. For instance, five years at a highly funded, top-tier school are treated the exact same by our model as five years at an under-resourced school, which likely skews the true economic return of that education.

Finally, our results are tied to this specific subsample of Malaysian workers. While the trends we uncovered, especially the role of secondary school as an equalizer, are striking, we have to be careful not to generalize these findings too broadly to other populations or different time periods without further testing.

References

[Wooldridge, 2019] Wooldridge, J. M. (2019). *Introductory Econometrics: A Modern Approach*. South-Western, Cengage Learning.